Lecture 28
Comparing two means

Review

1 What is the pooled proportion?

$\hat{p} = (X_1 + X_2)/(n_1 + n_2)$: all the successes over all the observations in both samples

2 What are independent samples?

one sample tells us nothing about the other: separate random samples, or randomized groups

3 What are dependent samples?

each value is paired with one in the other sample: the same subjects twice, twins

Review: to pool or not? (two proportions)

null hypothesis or goal	the SE we use	pool?
$H_0: p_1 = p_2$	$\text{SE}_0 = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}, \quad \hat{p} = \frac{X_1 + X_2}{n_1 + n_2}$	**pooled**
$H_0: p_1 - p_2 = d_0,\ d_0 \neq 0$	$\text{SE} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	**unpooled**
a $100(1-\alpha)\%$ CI for $p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) \pm Z_{1-\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	**unpooled**

pool only when H_0 says $p_1 = p_2$: then there is one p, and both samples estimate it

Two means: the notation

populations 1 and 2

the two groups we compare (for example, girls and boys)

$\mu_1,\ \mu_2$

the population means: the parameters

$\sigma_1,\ \sigma_2$

the population standard deviations: almost always unknown

$n_1, \overline{x}_1, s_1;\ \ n_2, \overline{x}_2, s_2$

the size, mean and standard deviation of each sample

$\overline{x}_1 - \overline{x}_2$

the estimate: $\overline{x}_1 - \overline{x}_2$ estimates $\mu_1 - \mu_2$

Two means: the hypotheses

$H_0: \mu_1 = \mu_2$

the same as $H_0: \mu_1 - \mu_2 = 0$: no difference between the populations

$H_A: \mu_1 - \mu_2 > 0$

upper-tailed: population 1 has the larger mean

$H_A: \mu_1 - \mu_2 < 0$

lower-tailed: population 1 has the smaller mean

$H_A: \mu_1 - \mu_2 \neq 0$

two-sided: a difference in either direction

The sampling distribution of $\bar{x}_1 - \bar{x}_2$

its center

$\mu_1 - \mu_2$: **on average, the estimate $\overline{x}_1 - \overline{x}_2$ hits the true difference**

its SE: variances add for independent samples

$$\text{SE} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

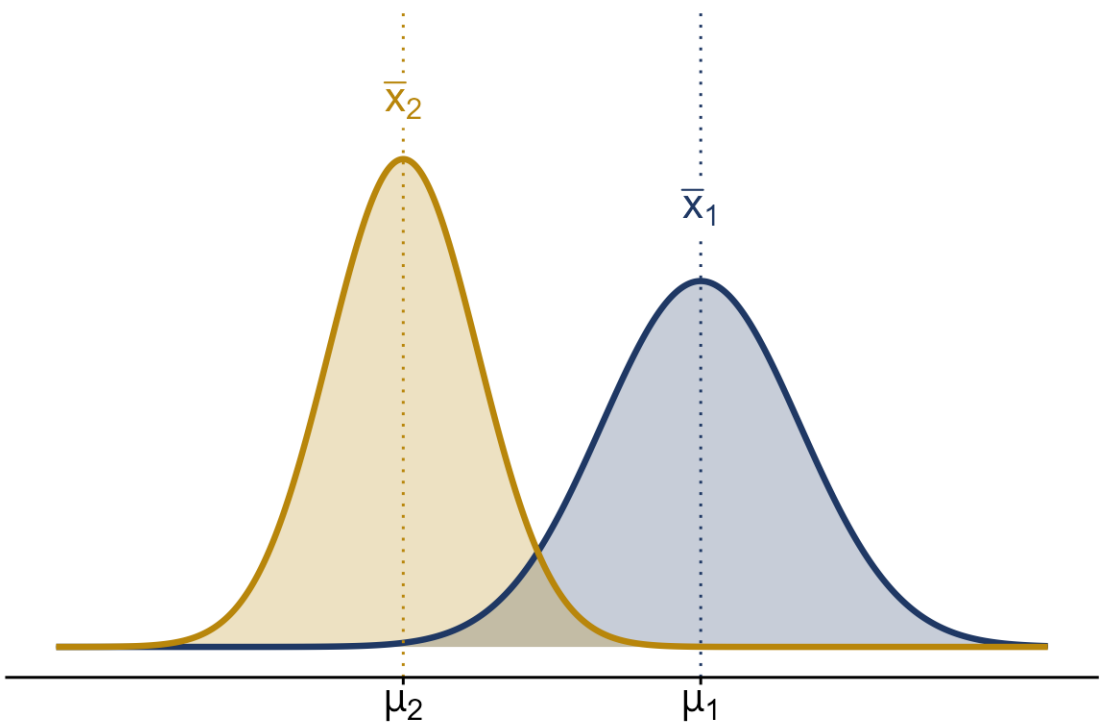
its shape

normal if both populations are normal; approximately normal if both $n \geq 30$

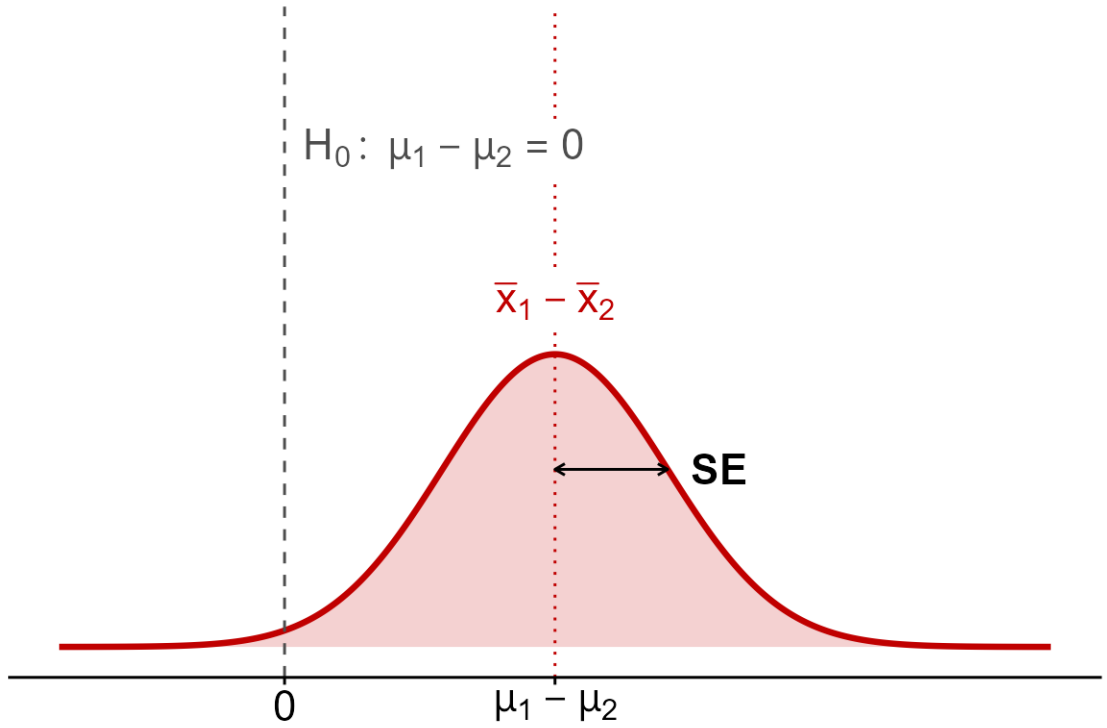
its distribution: approximately, once s_1, s_2 replace σ_1, σ_2 in the SE

$$\frac{(\overline{x}_1 - \overline{x}_2) - (\mu_1 - \mu_2)}{\text{SE}} \sim t(\min(n_1 - 1, n_2 - 1))$$

Picturing $\bar{x}_1 - \bar{x}_2$



two sample means, each with its own SE



the difference: wider, because variances add

$\bar{x}_1 - \bar{x}_2$ is centred at $\mu_1 - \mu_2$, and more spread out than either mean: the variances add

The two independent samples t-test

σ_1, σ_2 unknown

use s_1, s_2: $\mathrm{SE} = \sqrt{s_1^2/n_1 + s_2^2/n_2}$

the test statistic

$$t_{obs} = \frac{(\overline{x}_1 - \overline{x}_2) - 0}{\mathrm{SE}}$$

degrees of freedom

$df \approx \min(n_1 - 1, n_2 - 1)$: **an approximation (next slide)**

the P-value

the t-table with that df: the tail(s) H_A points to

assumptions

two independent random samples; both populations normal, or both $n \geq 30$

Degrees of freedom: an approximation

the Welch df (what software uses), with $v_1 = s_1^2/n_1,\ v_2 = s_2^2/n_2$

$$df = \frac{(v_1 + v_2)^2}{v_1^2/(n_1 - 1) + v_2^2/(n_2 - 1)}$$

in practice, by hand

$df \approx \min(n_1 - 1, n_2 - 1)$: **a quick approximation to the Welch df**

why that is safe

it is never larger than the Welch df, so our P-values come out a little too big: conservative

either way, an approximation

t_{obs} **only approximately follows a t distribution, even with the Welch df**

Two means: to pool or not?

test or interval	the SE (and df) we use	pool?
t-test, assume $\sigma_1 = \sigma_2$	$\text{SE} = s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}, \quad df = n_1 + n_2 - 2$	**pooled**
t-test, σ's may differ	$\text{SE} = \sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}, \quad df \approx \min(n_1 - 1, n_2 - 1)$	**unpooled**
$100(1-\alpha)\%$ CI, $\sigma_1 = \sigma_2$	$(\overline{x}_1 - \overline{x}_2) \pm t_{1-\alpha/2}\, s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}, \quad df = n_1 + n_2 - 2$	**pooled**
$100(1-\alpha)\%$ CI, may differ	$(\overline{x}_1 - \overline{x}_2) \pm t_{1-\alpha/2}\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}, \quad df \approx \min(n_1 - 1, n_2 - 1)$	**unpooled**

$s_p^2 = ((n_1 - 1)s_1^2 + (n_2 - 1)s_2^2)/(n_1 + n_2 - 2)$. **Default: unpooled ($H_0$ says nothing about σ)**

Example: girls, boys and sports

Girls and boys aged 7 to 11 are believed to spend the same average time each day playing sports. A study took a random sample of each; both populations are normal. Is there a difference in the mean time girls and boys spend playing sports each day? (OpenStax, Example 10.1)

girls: $n = 9,\ \overline{x} = 2$ **hours,** $s = 0.866$**;** **boys:** $n = 16,\ \overline{x} = 3.2$ **hours,** $s = 1.00$**;** $\alpha = 0.05$

(a) State H_0 and H_A. Which tail?

(b) Find the SE and t_{obs}. What is the df?

(c) Find the P-value from the t-table. What is the decision?

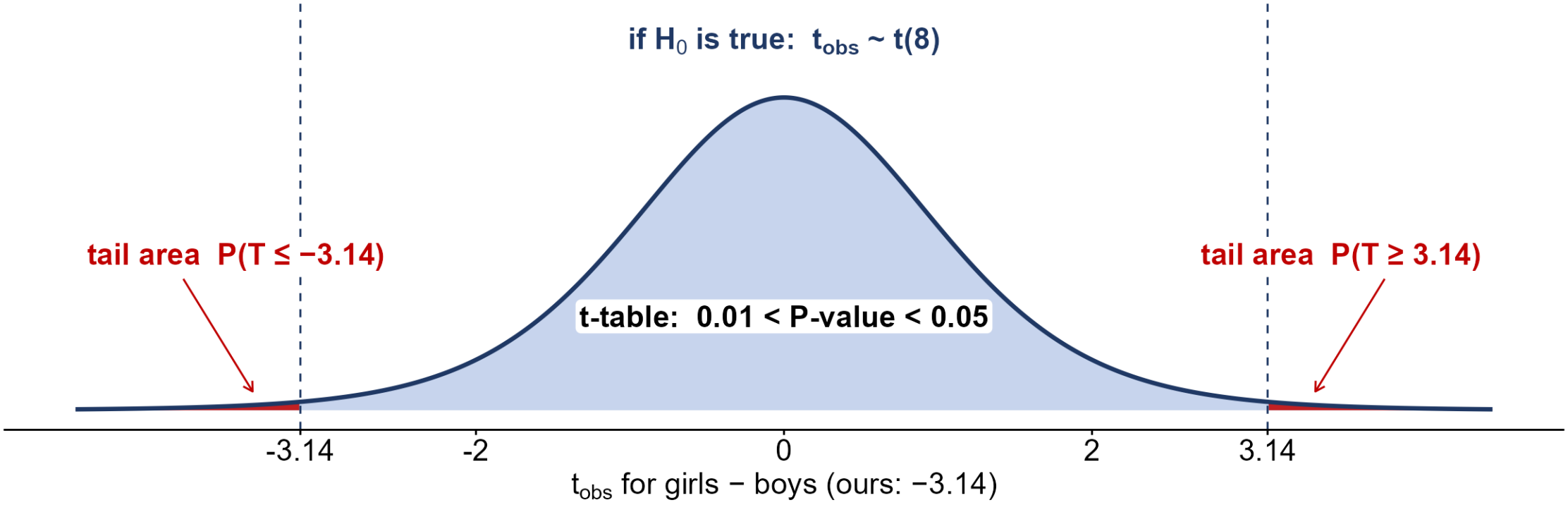
(a) $H_0: \mu_g - \mu_b = 0$ $H_A: \mu_g - \mu_b \neq 0$ **(two-sided)**

(b) $\text{SE} = \sqrt{0.866^2/9 + 1.00^2/16} = 0.3819;\ df = \min(8,15) = 8$

$t_{obs} = (2 - 3.2)/0.3819 = -3.14$

(c) $2.306 < 3.14 < 3.355:\ 0.01 <$ **P-value** $< 0.05:$ **reject** H_0

The sports P-value



$0.01 <$ **P-value** < 0.05 **(software: 0.0138): girls and boys differ in mean time playing sports**

Practice: who keeps workers longer?

A study asks whether Company A keeps its workers longer than Company B. Random samples of workers from each company give each worker's years with the company; both populations are normal. Does Company A keep its workers longer, on average? (OpenStax, Try It 10.2)

A: $n = 15,\ \overline{x} = 5$ **years,** $s = 1.2$**;** **B:** $n = 20,\ \overline{x} = 4.5$ **years,** $s = 0.8$**;** $\alpha = 0.05$

(a) State H_0 and H_A. Which tail?

(b) Find the SE and t_{obs}. What is the df?

(c) Find the P-value from the t-table. What is the decision?

(a) $H_0: \mu_A - \mu_B = 0$ $H_A: \mu_A - \mu_B > 0$ **(upper-tailed)**

(b) $\text{SE} = \sqrt{1.2^2/15 + 0.8^2/20} = 0.3578;\ df = \min(14,19) = 14$

$t_{obs} = 0.5/0.3578 = 1.40$

(c) $1.345 < 1.40 < 1.523:\ 0.075 <$ **P-value** < 0.10**, above 0.05: do not reject** H_0

Example: pooling for girls and boys

Back to girls, boys and sports. The two sample SDs are close (0.866 and 1.00), so assuming $\sigma_g = \sigma_b$ is reasonable here. Redo the test with the pooled SD.

girls: $n = 9,\ s = 0.866$; **boys:** $n = 16,\ s = 1.00$; $\overline{x}_g - \overline{x}_b = -1.2$; $\alpha = 0.05$

(a) Find s_p and the pooled SE.

(b) Find t_{obs} and the df. Bracket the P-value.

(c) Compare with the unpooled test. What did pooling buy us?

(a) $s_p = \sqrt{(8(0.866^2) + 15(1.00^2))/23} = 0.9555;\ \text{SE} = 0.9555\sqrt{1/9 + 1/16} = 0.3981$

(b) $t_{obs} = -1.2/0.3981 = -3.01;\ df = 9 + 16 - 2 = 23$: $0.005 <$ **P-value** < 0.01

(c) unpooled: $df = 8,\ 0.01 <$ **P-value** < 0.05. **Pooling: more df, a smaller P-value (more power)**

Pooling: advantages and disadvantages

when it is reasonable

similar sample SDs (larger / smaller under about 2), similar sample sizes

the advantage

more df ($n_1 + n_2 - 2$) and an exact t distribution: more power if $\sigma_1 = \sigma_2$

the disadvantage

if $\sigma_1 \neq \sigma_2$ and the n's differ, the Type I error rate is not α

for example

$n_1 = 10, \sigma_1 = 3$ **vs** $n_2 = 40, \sigma_2 = 1$**: a 5% pooled test rejects a true H_0 26% of the time**

The rank-sum test (non-parametric)

Wilcoxon rank-sum test

compares two independent samples without assuming normal populations

how it works

rank all $n_1 + n_2$ values together, then compare the two samples' rank sums

when to use it

small samples from skewed populations, outliers, or ranked (ordinal) data

more

Resources → Extras → Non-parametric Tests (also the signed-rank test for pairs)

Matched pairs: the idea

matched pairs

each value in one sample is paired with one in the other: same subject twice

the differences

$d = x_1 - x_2$ **for each pair: the n differences are the data**

the parameter

μ_d: **the mean difference in the population of pairs**

the big idea

a two-sample design becomes a one-sample t-test on the differences

The matched-pairs t-test

$H_0: \mu_d = 0$

no difference within pairs, on average

H_A

$\mu_d > 0$ **(upper),** $\mu_d < 0$ **(lower),** $\mu_d \neq 0$ **(two-sided)**

the test statistic

$t_{obs} = (\bar{d} - 0)/(s_d/\sqrt{n})$: $\bar{d}, s_d$ **= the mean and SD of the differences**

degrees of freedom

$df = n - 1$, **where n is the number of pairs**

assumptions

a random sample of pairs; the differences are normal, or $n \geq 30$

Data: hypnotism and pain

Subject	A	B	C	D	E	F	G	H
Before	6.6	6.5	9.0	10.3	11.3	8.1	6.3	11.6
After	6.8	2.4	7.4	8.5	8.1	6.1	3.4	2.0
d = after − before	0.2	−4.1	−1.6	−1.8	−3.2	−2.0	−2.9	−9.6

Randomly selected subjects had a sensory pain score measured before and after hypnotism; a lower score means less pain. X = a subject's score before, Y = the same subject's score after, $d = Y - X$. $n = 8$ pairs. (OpenStax, Example 10.11)

Example: does hypnotism reduce pain?

A study asked whether hypnotism reduces pain. Each subject's pain score was measured before and after hypnotism (lower means less pain), and the differences are normal. Are the scores lower after hypnotism, on average? (OpenStax, Example 10.11)

$n = 8$ **subjects;** d **= after – before;** $\overline{d} = -3.125,\ s_d = 2.91$**;** $\alpha = 0.05$

(a) Why matched pairs? State H_0 and H_A. Which tail?

(b) Find the SE and t_{obs}. What is the df?

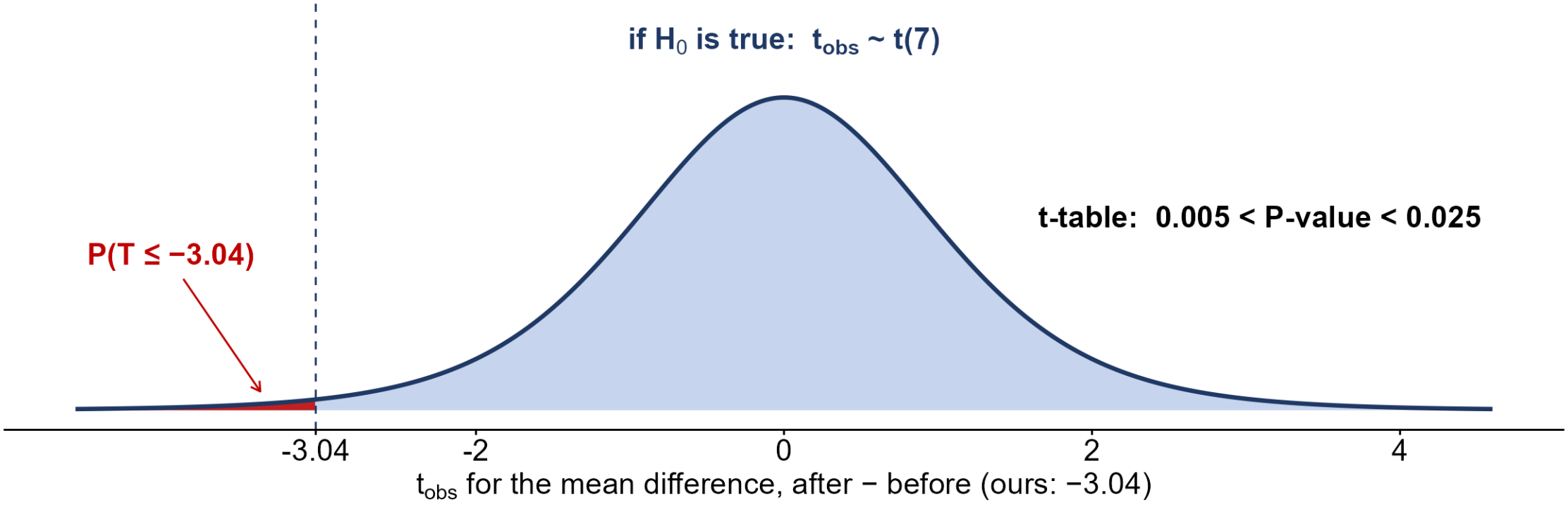
(c) Find the P-value from the t-table. What is the decision?

(a) the same 8 subjects, twice; $H_0: \mu_d = 0$ $H_A: \mu_d < 0$ **(lower-tailed)**

(b) $\text{SE} = 2.91/\sqrt{8} = 1.0288;\ t_{obs} = -3.125/1.0288 = -3.04;\ df = 8 - 1 = 7$

(c) $2.365 < 3.04 < 3.499$**:** $0.005 <$ **P-value** < 0.025**, below 0.05: reject** H_0

The hypnotism P-value



$0.005 <$ **P-value** < 0.025 **(software: 0.0094): pain scores are lower after hypnotism, on average**

Why pair?

Suppose we forgot the pairing and treated the hypnotism scores as two independent samples of 8.

before: $\bar{x} = 8.71,\ s = 2.18$; **after:** $\bar{x} = 5.59,\ s = 2.61$; $\bar{x}_{\text{after}} - \bar{x}_{\text{before}} = -3.125$

(a) Find the unpooled SE and t_{obs}.

(b) Compare with the matched-pairs test. Why is the paired SE smaller?

(a) $\text{SE} = \sqrt{2.18^2/8 + 2.61^2/8} = 1.2023;\ \ t_{obs} = -3.125/1.2023 = -2.60$

(b) paired $\text{SE} = 1.0288 < 1.2023$**: each subject is their own control**

Data: shot-put with each hand

Student	1	2	3	4	5	6	7
Dominant hand	30	26	34	17	19	26	20
Weaker hand	28	14	27	18	17	26	16
d = dominant − weaker	2	12	7	−1	2	0	4

Seven eighth graders pushed the shot-put with their dominant (writing) hand and with their weaker hand, in feet. X = the dominant-hand distance, Y = the weaker-hand distance, $d = X - Y$. $n = 7$ pairs. (OpenStax, Example 10.13)

Practice: shot-put with each hand

Seven eighth graders thought they could push the shot-put equal distances with either hand. Each pushed with their dominant hand and with their weaker hand; the differences are normal. Is there a mean difference between the two hands? (OpenStax, Example 10.13)

$n = 7$ **students;** d **= dominant – weaker (feet);** $\bar{d} = 3.71,\ s_d = 4.50$**;** $\alpha = 0.05$

(a) Why matched pairs? State H_0 and H_A. Which tail?

(b) Find the SE and t_{obs}. What is the df?

(c) Find the P-value from the t-table. What is the decision?

(a) the same 7 students, both hands; $H_0: \mu_d = 0$ $H_A: \mu_d \neq 0$ **(two-sided)**

(b) $\text{SE} = 4.50/\sqrt{7} = 1.7008;\ t_{obs} = 3.71/1.7008 = 2.18;\ df = 7 - 1 = 6$

(c) $1.943 < 2.18 < 2.447$**:** $0.05 <$ **P-value** < 0.10**, above 0.05: do not reject** H_0

Practice: which test?

1 **Mean nightly sleep: random samples of 40 first-years and 40 seniors**

two independent samples t-test: two separate samples, a quantitative response

2 **Reaction time of 25 drivers, rested and after a sleepless night**

matched-pairs t-test: the same 25 drivers, measured twice

3 **Share who exercise daily: random samples of 200 dog and 200 cat owners**

two-proportion z-test: two independent samples, a yes/no response